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Faculté des Mathématiques et Informatique
Département D'Informatique

Machine Learning Course

By Dr. Asmaa Ourdighi

Lecture 1

Chapter 1: What is Machine Learning?

1. Definition and Principles.
2. Different Types of Learning.

What is Machine Learning?

Definition 1

- Arthur Samuel, one of the pioneers of the field, defined ML back in 1959 as: "**The field of study that gives computers the ability to learn without being explicitly programmed**."

This simple definition captures the essence of machine learning: the *computer doesn't follow a rigid script; it develops its own logic to solve a problem.*

- Imagine that you want to train a computer to predi
  developing diabetes based on their medical records.

With traditional programming, you would need to manual
as:

→ "***If*** *the patient's BMI (*Body Mass Index) *is certain threshold, then the risk is elevated*."

→"***If*** *there is a family history of diabetes, incre*

,,,,

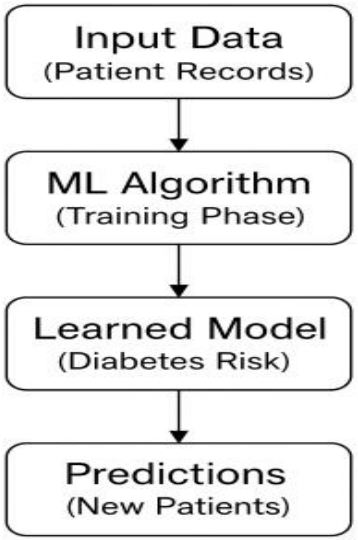
Warning!!!!

This approach quickly becomes unmanageable,

- Noisy
- full of exceptions.
- Real-world medical data is complex
- . .

What is Machine Learning?

With Machine Learning, however, you do not write these rules yourself. Instead, you **provide the algorithm** with **a large dataset** of patient records, some labeled as "*diabetic*" and others as "*non-diabetic.*" The system then analyzes these data and automatically builds **a mathematical model** that determines which **features**, such as *glucose levels, age, weight, lifestyle habits*…, are most relevant for predicting diabetes risk,



What is Machine Learning?

Definition 2

- A more formal and widely accepted definition was later provided by Tom Mitchell in 1997. It helps to better understand what a learning process is from a computational perspective: "**A computer program is said to learn from experience E with respect to some class of tasks T and a performance measure P, if its performance at tasks in T, as measured by P, improves with experience E**.".

•*To illustrate this definition, let's use a concrete example:* ***spam detection****.*

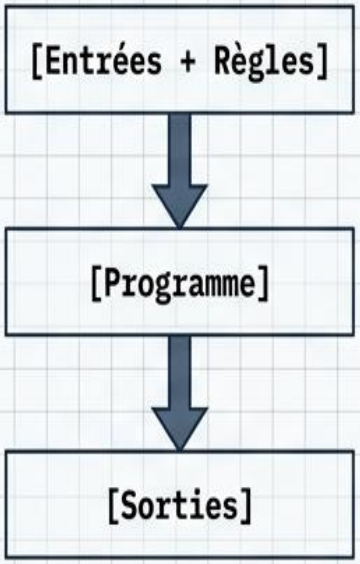
→ *Task (T): Predicting whether an email is spam or not.*

→ *Experience (E): Training the program on a dataset of already classified emails (labeled as "spam" or "not spam"). The more emails the program analyzes, the more its experience grows.*

→ *Performance Measure (P): The accuracy of the classification on new emails. For example, if the program correctly classifies 95% of new emails, its performance is 95%.*

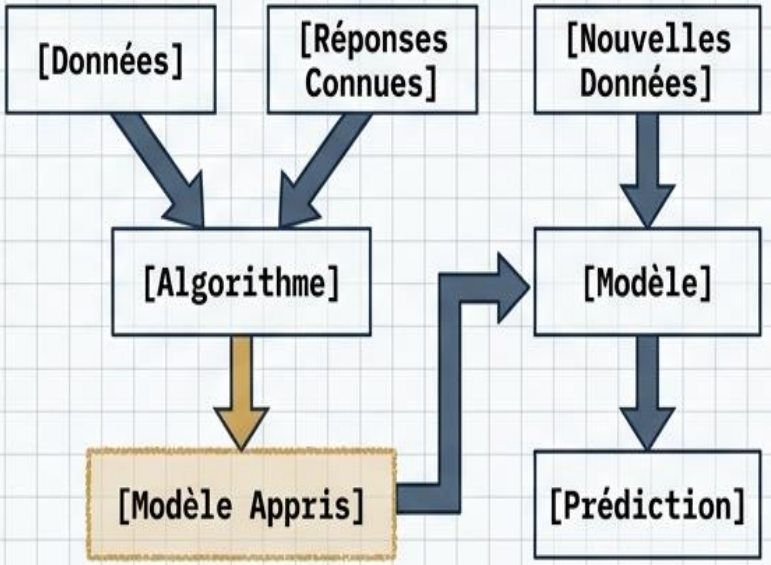
Le Changement de Paradigme

Programmation Traditionnelle



Exemple : Calcul de la TVA (Prix -> Règle -> Résultat)

Machine Learning



Pourquoi le ML ? Pour des tâches comme la reconnaissance faciale, les règles explicites (luminosité, orientation, position, expression, arrière-plan, résolution) sont infinies.

Axiome : Le système n'exécute plus des règles codées manuellement ; il déduit des régularités statistiques à partir de l'expérience.

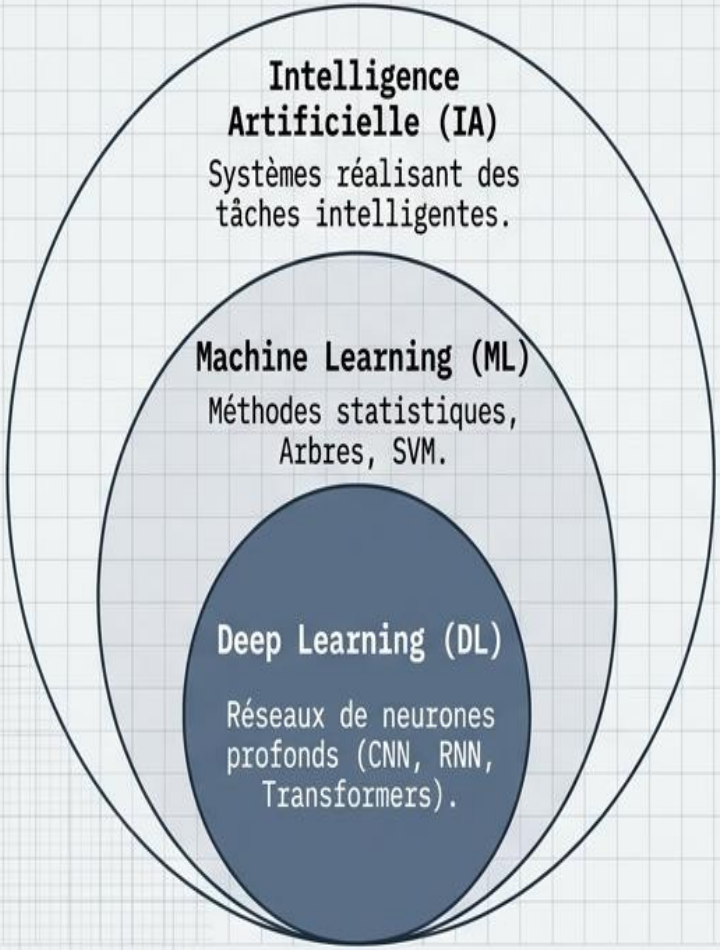
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- **A key branch of AI that enables systems to learn from data**
  Replaces fixed rules with adaptive algorithms
  Discovers patterns, predicts outcomes, and generalizes to new cases

L'Écosystème de l'IA : Topologie et Outils



Frameworks ML :

- Scikit-Learn
- Pandas

Frameworks DL :

- TensorFlow
- PyTorch

Calcul & Data :

- NumPy
- Hadoop
- Spark

Infrastructure :

- GPU/TPU
- AWS SageMaker
- Google Vertex
- Azure ML

⚠ **Axiome** fondamental de l'analyse : Corrélation ≠ Causalité.

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The machine learning process is a systematic journey that transforms raw data into actionable intelligence. It begins with

- **"Data Preparation"**, a critical phase involving the **gathering of relevant data**. This data is then **meticulously cleaned to handle inconsistencies** and **missing values, preprocessed into a suitable format for the chosen algorithm**, and finally **split into training, validation, and testing sets**.
- Next, in Model **"Selection & Training",** a **suitable machine learning model** is chosen and trained on the prepared data. The **model learns by iteratively adjusting** its internal parameters **to minimize a defined loss function**.

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- The trained model's performance is then rigorously assessed in the **"Model Evaluation"** step using **unseen test data** and **relevant metrics**. If the results are unsatisfactory, this stage often requires **fine-tuning hyperparameters** or **even revisiting earlier phases like data preparation** to improve performance
- Finally, a successful model is deployed into a real-world application or system during the **"Deployment & Monitoring"** phase. Once deployed, the model's performance should be continuously monitored and, when necessary, it can be retrained with new data to maintain its accuracy and relevance.

Pipeline Machine Learning



- *Le processus est itératif : en cas de résultats insuffisants, on revient aux phases antérieures*

Features et Labels

Feature (Caractéristique)

Variable utilisée comme entrée du modèle

Exemple pour une maison :
- Surface
- Nombre de chambres
- Localisation
- Âge du bâtiment

Label (Étiquette/Cible)

La réponse qu'on cherche à prédire

Exemple :
- Regression: Prix de la maison

- Classification: Chat vs Chien vs Cheval

Exemple : Prédiction de Prix Immobilier

Données d'entraînement :

Surface (m²)	Chambres	Localisation	**Prix (M)**
80	2	Centre	**12M**
120	3	Centre	**18M**
200	4	Banlieue	**20M**
150	3	Centre	**22M**

Le modèle apprend à relier Features (surface, chambres, localisation) à Label (prix)

🧠 Modèle et Paramètres

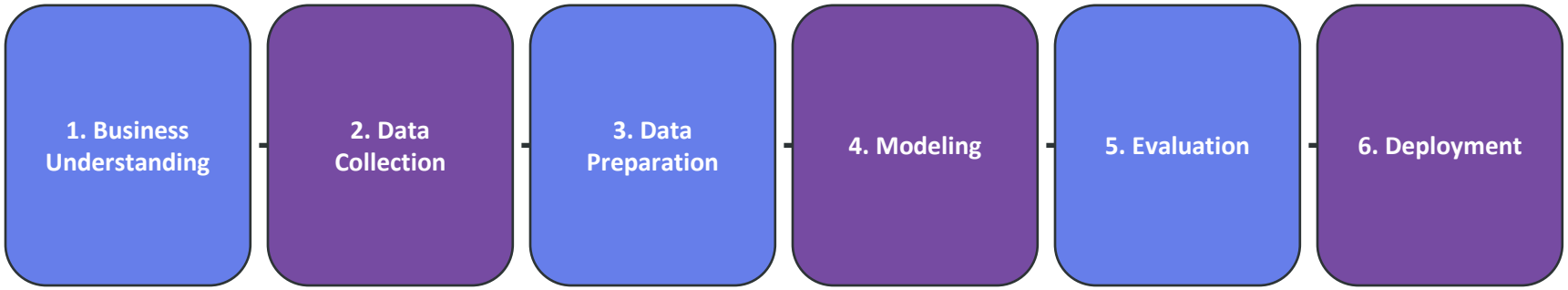
Le modèle est la représentation apprise

$$\hat{y} = f_\theta(x)$$

x = données entrée | θ = paramètres | ŷ = prédiction

- Objectif : apprendre les paramètres θ qui donnent les meilleures prédictions
- Les paramètres sont ajustés pendant l'entraînement **via un algorithme d'optimisation**

Cycle de Vie d'un Projet ML



- **PROCESSUS ITÉRATIF : en cas de résultats insuffisants, revenir aux phases antérieures**

Supervised Learning

Supervised learning is one of the **most widely studied** and applied paradigms in machine learning. It relies on the existence of a **labeled dataset**, where each input observation is associated with an expected output. These labels act as a "**guide**" for the algorithm, enabling it to learn the correspondence between inputs and outputs in order to make reliable predictions on new, unseen data.

- **Definition**

  Supervised learning refers to the set of machine learning methods that aim to ***approximate an unknown function mapping an input space X to an output space Y.*** *We are given a set of pairs (x^i, y^i), where $x^i \in X$ represents* ***an observation*** *and $y^i \in Y$ its* ***corresponding label****. These labels are assumed to come from a* ***hidden function*** *$\emptyset: \boldsymbol{X \to Y}$, affected by some random* ***noise ε****.* ***The goal of supervised learning is to build a model $f: X \to Y$ such that, for every $x \in X$, $f(x) \approx \emptyset(x)$.***

- Beyond this mathematical formulation, supervised learning is commonly applied to two major types of problems:
  - ✓ **Classification**, where the **output corresponds to a discrete category**.
    For example, determining whether an email is "spam" or "not spam," recognizing handwritten digits from images, or predicting the presence or absence of a disease from medical data.
  - ✓ **Regression**, where the **output is a continuous value**.
    Examples include predicting the price of a house based on its characteristics (such as size, location, and number of rooms) or estimating a building's energy consumption depending on weather conditions and usage patterns.
- In both cases, the **quality of supervised learning** depends not only on **the relevance of the training data** but also on **the generalization ability of the learned model**. The objective is not to memorize the training examples but to learn a sufficiently robust mapping that can provide accurate predictions on previously unseen data.

Unsupervised Learning

Unsupervised learning is **another fundamental branch** of machine learning. Unlike supervised learning, it does **not rely on labeled data**: only observations are available, without any prior indication of the expected output. The objective is to automatically **discover hidden patterns, structures, or groupings in the data**, allowing the algorithm to learn by itself without direct supervision.

- **Definition**

  Unsupervised learning ***refers to a set of methods*** *which, given a collection of observations* $\{x^1, x^2, \dots, x^n\} \in X$*, aim to* ***reveal regularities or latent structures without access to associated labels****. In other words, the algorithm attempts to determine a function* $g: X \to S$ *that organizes, clusters, or reduces the data based on similarities, where* S *denotes the space of discovered structures or clusters.*

- The most common tasks include:
  - ✓ **Clustering**, where data points are grouped into homogeneous classes, for example grouping customers based on their purchasing behavior.
  - ✓ **Dimensionality reduction**, such as PCA (Principal Component Analysis), which projects data into a lower-dimensional space while preserving essential information, for visualization or compression.
  - ✓ **Anomaly detection**, which identifies unusual data points, highly relevant in cybersecurity to detect suspicious network activity.

Examples of applications; Automatic organization of image libraries based on visual similarity, Product recommendation systems built from user profiles without explicit labels, Fraud or intrusion detection in systems where no prior labeled examples are provided.

Semi-Supervised Learning

Semi-supervised learning lies **between supervised and unsupervised** paradigms. It leverages **both labeled and unlabeled data**, making it especially valuable when acquiring large volumes of raw data is easy, but labeling them is costly, time-consuming, or requires expert knowledge. The central idea is to **exploit the abundance of unlabeled** data to improve the **learning process guided by a smaller set of labeled examples**.

- **Definition**

  Semi-supervised learning consists of algorithms ***designed to infer a predictive*** *function* $f: X \to Y$ *by* ***combining a small set of labeled data*** $\{(x^i, y^i)\}$ *with* ***a much larger set of unlabeled data*** $\{x^j\}$*. The presence of unlabeled observations allows the algorithm to better capture the structure of the input space, which improves generalization.*

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2. 1 Semi-Supervised Learning

- Semi-supervised learning **offers significant advantages**, particularly **regarding data cost and quality**. Its main strength is the reduction in the **cost of manual annotation**, since only a fraction of the data needs to be labeled. This is especially **relevant for large datasets** where labeling is a long and expensive process. Moreover, by being less dependent on human labels, this approach mitigates the risk of bias that could be introduced by annotators. Finally, using unlabeled data allows the model to explore the **underlying structure of the data**, which improves its performance compared to purely supervised approaches, especially when labeled samples are scarce.
- This method is used across a wide variety of fields:
  - ✓ **Image Classification:** In this area, only a small subset of images is manually annotated (e.g., as "dog" or "cat"), while the vast number of unlabeled images helps the model refine the decision boundaries between classes.
  - ✓ **Medical Diagnosis:** Experts carefully label a few disease cases, but the large amount of raw, unlabeled medical data (scans, X-rays) can be used to train a more robust model capable of recognizing complex patterns.
  - ✓ **Natural Language Processing (NLP):** Semi-supervised learning leverages a few annotated sentences (for sentiment analysis, for example) by combining them with large corpora of raw text to improve the accuracy of text classification models.

Reinforcement Learning

Reinforcement learning represents a distinct paradigm where **learning takes place through interaction with an environment**. Instead of being provided with explicit labels, the system learns by **performing actions and receiving feedback** in the form of rewards or penalties. The ultimate goal is to develop a policy that **maximizes the long-term cumulative reward**.

Definition

Reinforcement learning can be defined as a framework where an agent interacts with an environment by executing actions $a \in A$*, observes the resulting state* $s \in S$ *and receives a reward signal* r*. The task is to learn a policy* $\pi: S \rightarrow A$ *that maximizes the expected cumulative reward across time.*

- Reinforcement learning is **fundamentally based on the principle of exploration through trial and error**. Unlike supervised learning, where the correct outputs are explicitly provided, the **agent must experiment with different actions**, **observe the resulting outcomes**, and **iteratively refine its strategy**. This exploratory process allows the system to uncover novel solutions that would **not emerge from a strictly deterministic approach.**

- Another central feature lies in the **management of immediate versus delayed rewards**. **Some actions yield instant benefits, while others only prove advantageous in the long run**. Consequently, the agent ***must balance*** the trade-off between **seeking short-term gains** (exploitation) and anticipating potentially **greater future rewards** (exploration). This tension makes reinforcement learning particularly well-suited to complex environments where present decisions significantly affect long-term performance.
- Finally, reinforcement learning is characterized by **its adaptability to dynamic and uncertain environments**. Since the **conditions surrounding the agent may change unpredictably**, the system must **continuously adjust its policy in real time**. This adaptability is one of the major strengths of reinforcement learning, enabling autonomous systems to operate effectively in **evolving or unstable contexts**.

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2. 1 Reinforcement Learning

- This method is used across a wide variety of fields:
  - ✓ In **the domain of games**, reinforcement learning has led to groundbreaking achievements. Systems such as **AlphaGo** have surpassed the strongest human players in the game of Go, long considered a grand challenge for artificial intelligence. Similarly, in chess and various video games, reinforcement learning agents have demonstrated the ability to acquire innovative strategies that often surprise and surpass human expectations.
  - ✓ In **robotics**, reinforcement learning enables machines to acquire advanced skills without the need for explicit programming. Robots can learn to walk steadily, manipulate objects with precision, or navigate complex environments. This approach promotes the development of versatile robots capable of adapting their behavior to a wide range of tasks, thereby expanding their usefulness in real-world applications.
  - ✓ Another major field of application is **autonomous driving**. Self-driving vehicles must constantly make rapid, context-aware decisions in environments that are both uncertain and dynamic. Through reinforcement learning, these systems can optimize their driving strategies by balancing safety, energy efficiency, and passenger comfort, while continuously improving through interaction with real traffic conditions.

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QCM : Chapter 1: What is Machine Learning?

- **Q1.** Which of the following best captures the essence of *machine learning*?
  A) Programming explicit rules for every possible situation.
  B) Designing systems that improve automatically through experience.
  C) Feeding raw data into a computer without any algorithms.
  D) Using computers only for statistical analysis.
- **Q3.** Which of the following is an example of *unsupervised learning*?
  A) Predicting tomorrow's temperature based on historical data.
  B) Grouping customers into segments based on purchasing behavior.
  C) Training a robot to play chess by receiving rewards for good moves.
  D) Classifying handwritten digits into digits 0–9.
- **Q5.** Which of the following best illustrates reinforcement learning?
  A) A stock market prediction model trained on past labeled outcomes.
  B) A clustering algorithm grouping genes with similar expression patterns.
  C) An agent learning to maximize its score in a video game through trial and error.
  D) A supervised classifier detecting spam in emails.